1. INTRODUCTION

Automation has become essential in both industrial applications and daily life, offering increased efficiency and resource conservation. One area where automation can improve daily tasks is in gardening and plant care. Traditionally, plant watering is a manual task that can lead to overwatering or underwatering due to human error.

The Automated Plant Watering System aims to address this issue by ensuring plants receive the right amount of water at the right time. Using an Arduino UNO and a soil moisture sensor, the system monitors soil moisture levels. When the soil is dry, the system automatically activates a water pump to water the plant. Once the soil reaches the optimal moisture level, the pump turns off, preventing overwatering.

This system not only simplifies plant care but also promotes water conservation by using water only when needed. It's particularly beneficial for individuals with busy schedules, those with multiple plants, or areas facing water scarcity. With the added benefit of a real-time LCD display to show soil moisture levels and watering status, users can efficiently manage their plants' needs.

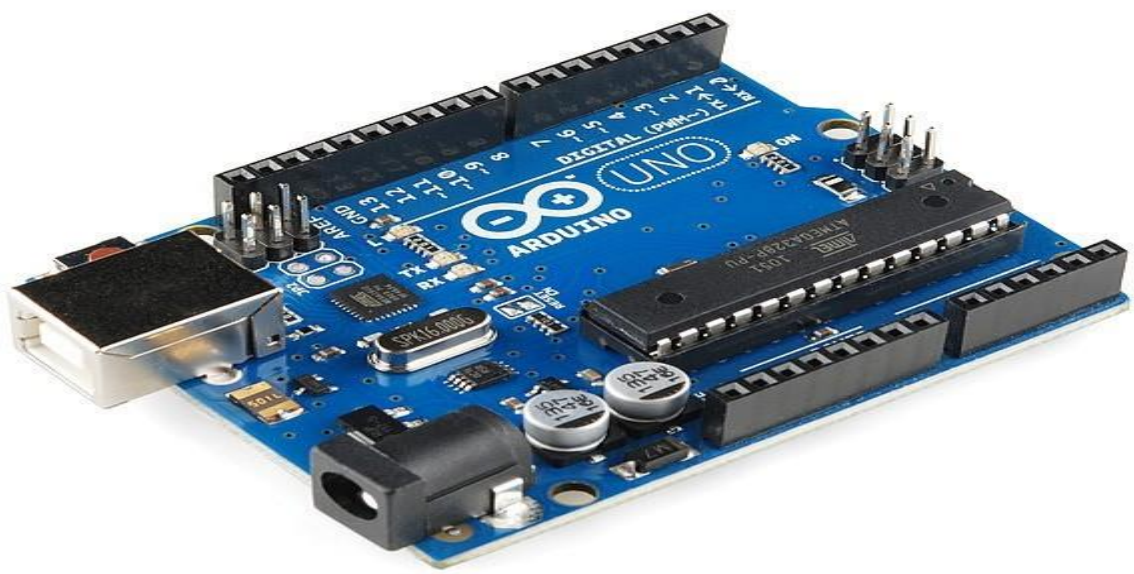
Applications:

1. Home Gardening – Ideal for individuals with houseplants
2. Indoor Plants – Perfect for keeping indoor plants hydrated without the need for constant attention.
3. Greenhouses – Helps maintain optimal soil moisture for various types of plants.

2. Hardware and Software requirements

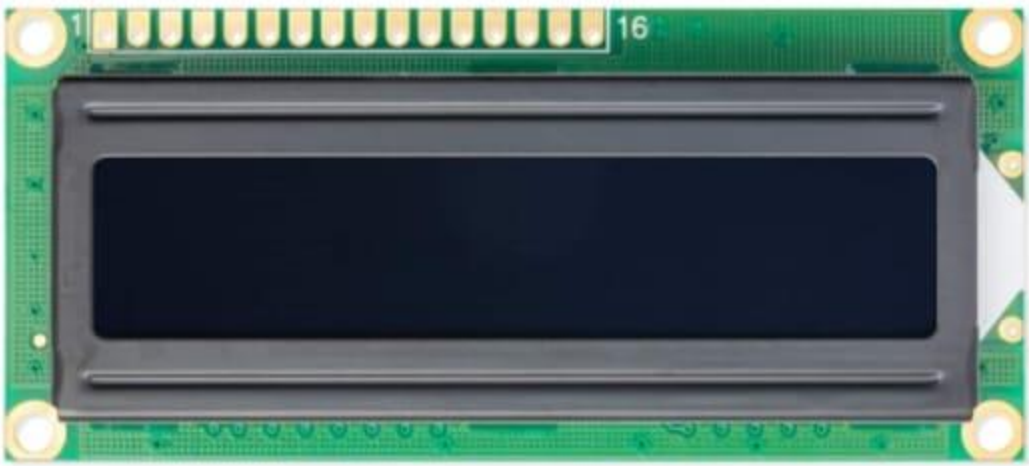
2.1 Hardware components

(a) Arduino UNO



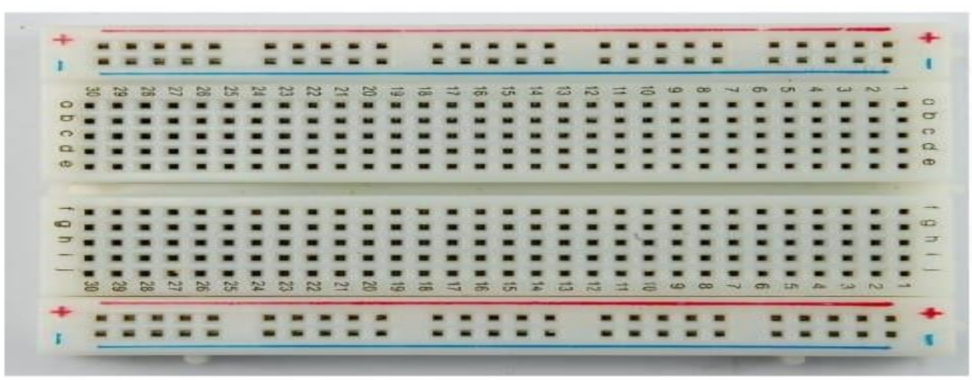
Arduino / Genuine Uno is a microcontroller board based on ATmega328P(datasheet).It has 14 digital input/output pins(of which 6 are used as PWM outputs),6 analog inputs, a 16 MHz quartz crystal a USB connection, a power jack, an ICSP header and a reset button. It contains everything needed to support the microcontroller; simply connect it to a computer with a USB cable or power it with a AC-to-DC adapter or battery to get started .You can tinker with your UNO without worrying too much about doing something wrong, worst case scenario you can replace the chip for a few dollars and start over again.

(b) LCD (Liquid Crystal Display)



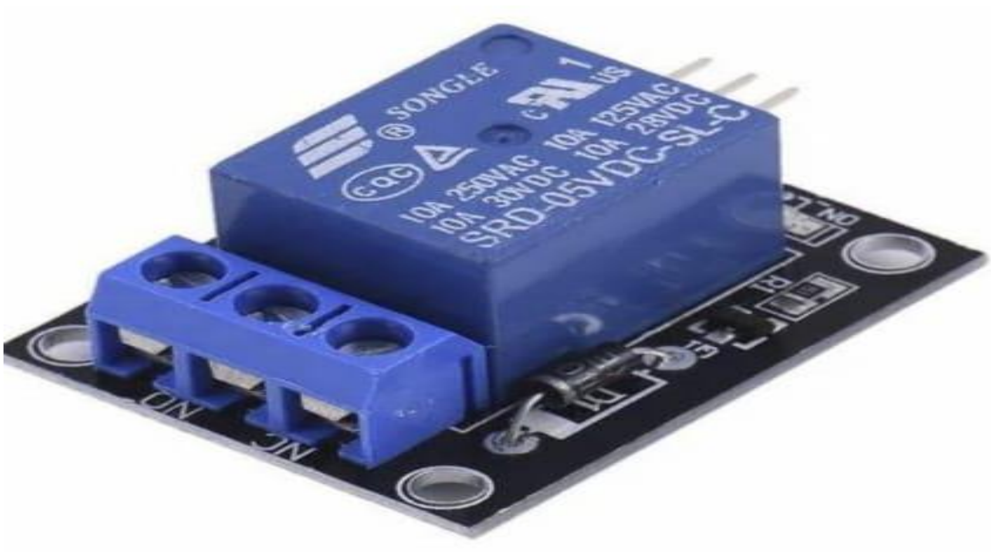
A liquid crystal display is a flat-panel display or their electrically modulated optical device that uses the light modulating properties of liquid crystals. Liquid crystals do not emit light directly, instead using a black light or reflector to produce in color or monochrome.

(c) Breadboard



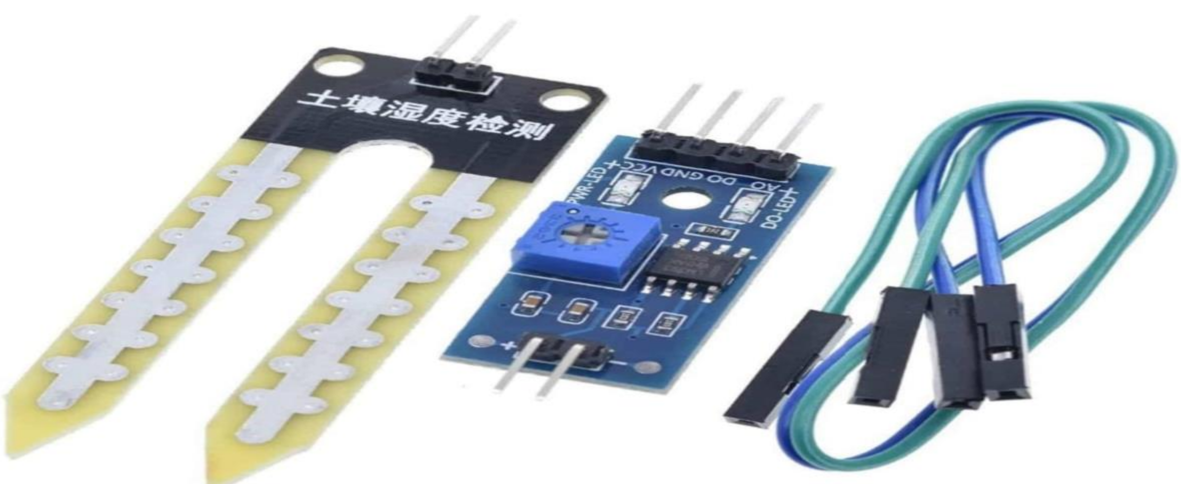
A breadboard is a solderless device for temporary prototype with electronics and test circuit designs.

(d)Relay



A relay is an electrically operated switch that allows a low-power control signal to switch a higher-power circuit.

(e) Soil moisture Sensor

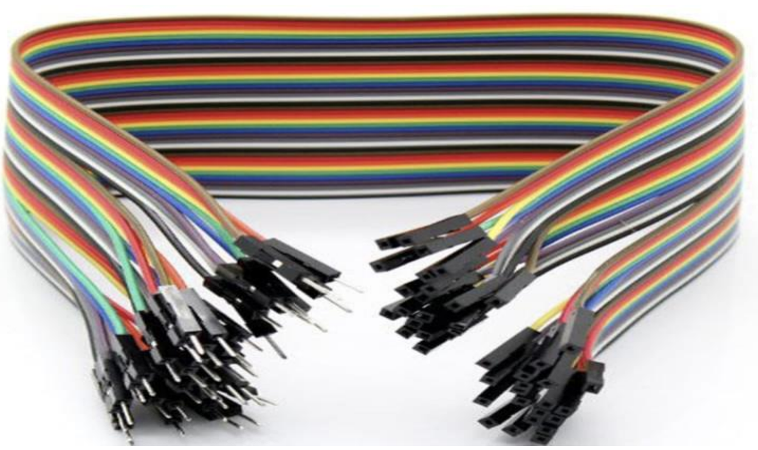


A soil moisture sensor measures the water content in the soil, helping determine when plants need watering.

(f)Water pump

A water pump is a device used to move water from one place to another. it is activated by the relay to deliver water to the plant when the soil moisture is low.

(g) Battery And Jumper wires



A battery provides the necessary power to operate the components

Jumper wires are flexible electrical wires used to make connections between components in a circuit.

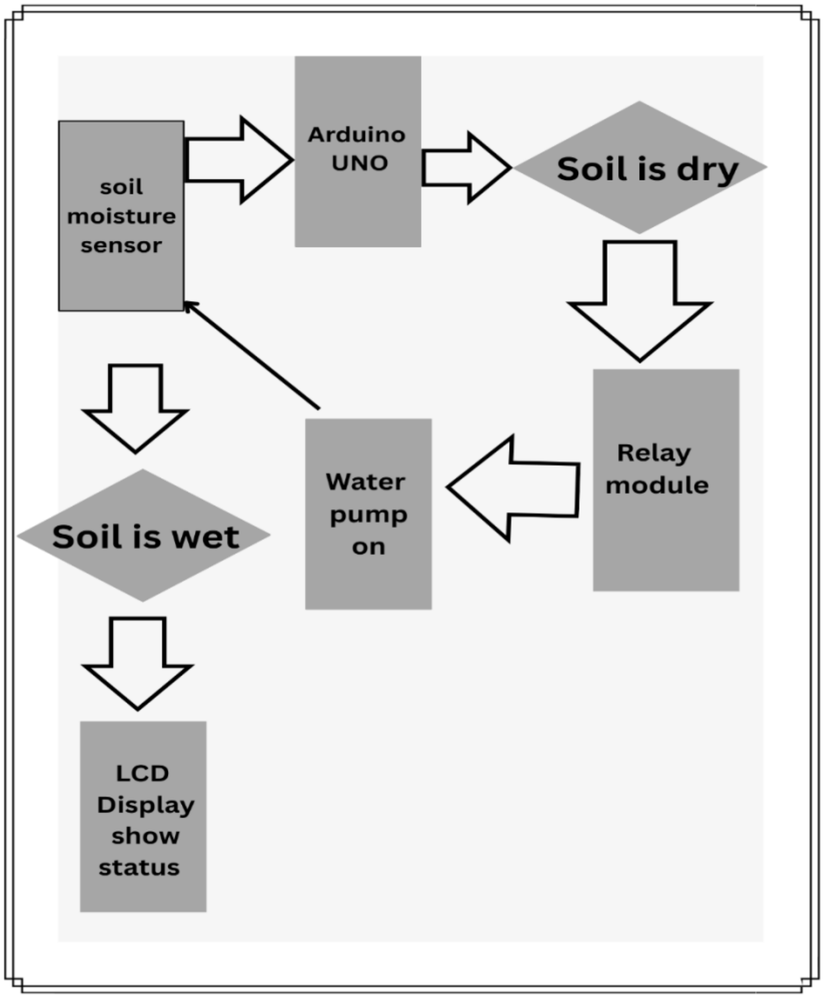
2.2 software requirements

(a) Arduino IDE Software

Arduino IDE software is an open source software to which a hobbyist can connect the AT mega chips. In this software the code can be written and uploaded to any AT mega chip and then the code can be executed on the chip. Many 3D printed electronics and Arduino-compatible use AT mega chip and hence the user can upload the program. Arduino can also be used firmware any electronics. Sketch is the window in which the program is to be written.

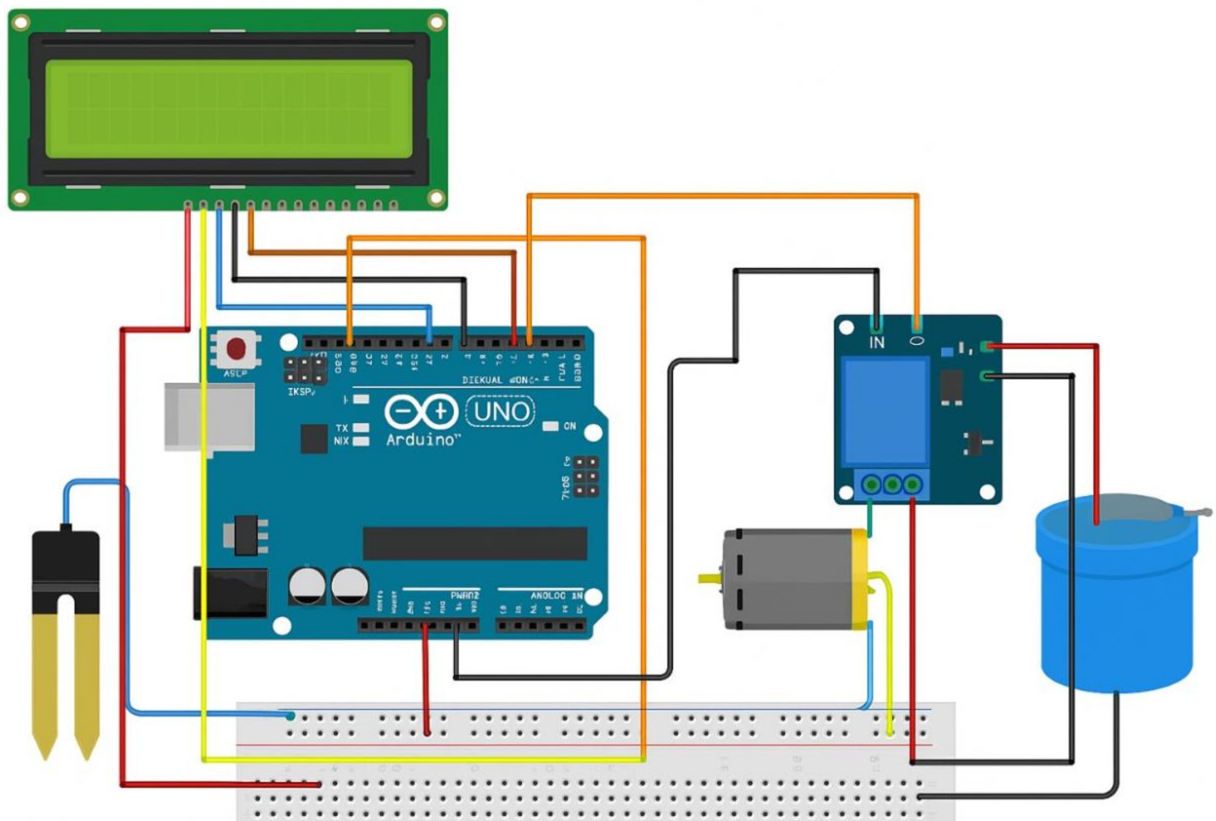
3.METHODOLOGY

3.1 Hardware Description:



The figure below shows the block diagram of the Automated Plant Watering System. It operates based on the moisture level present in the soil. When the soil becomes dry, the soil moisture sensor detects the low moisture content and sends a signal to the Arduino UNO. The Arduino then activates the water pump through the relay module to irrigate the plant. Once the soil moisture reaches the required level, the system automatically turns OFF the pump, ensuring optimal watering without human intervention.

CIRCUIT DIAGRAM



The principle behind the working of the Automated Plant Watering System is similar to basic automated control operations but is designed specifically for plant care. The soil moisture sensor continuously monitors the moisture level in the soil.

When the moisture level drops below a defined threshold, the microcontroller (Arduino UNO) identifies it as a signal to start watering. Upon detecting dry soil:

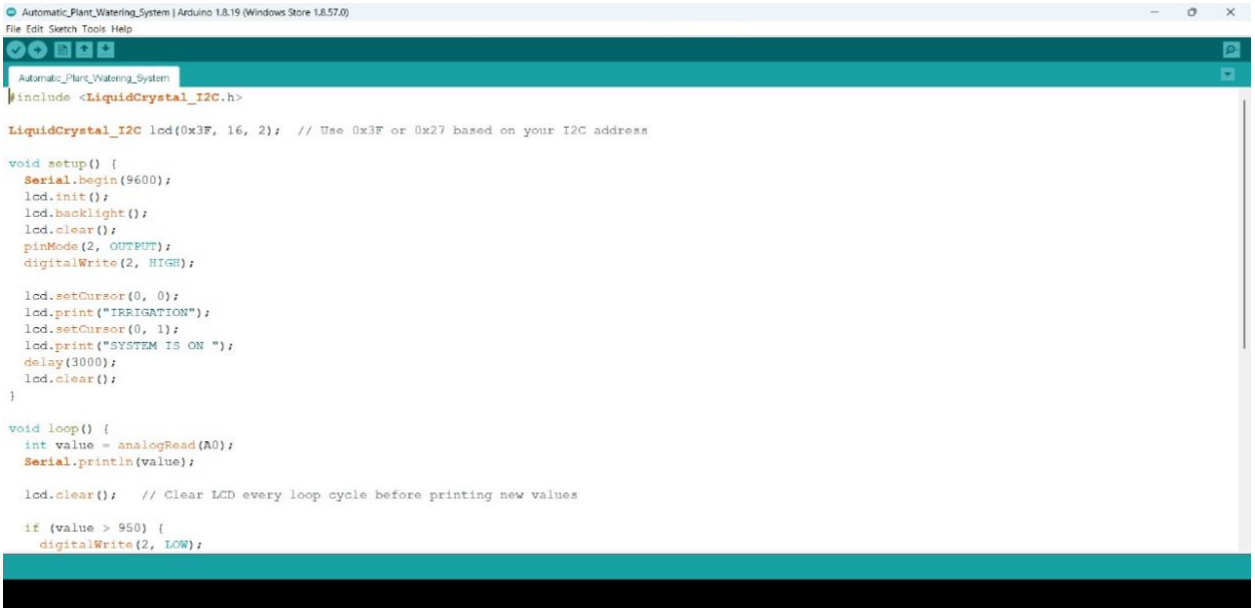
- The Arduino activates the relay module, which in turn switches ON the water pump.
- Water is supplied to the plant until the sensor detects that the soil has reached the required moisture level.
- During the process, the LCD display provides real-time updates such as "Soil is dry, watering...", "Watering complete", etc.

- If the soil moisture is already sufficient, the pump remains OFF and the system displays "Soil Moisture Normal" on the LCD.

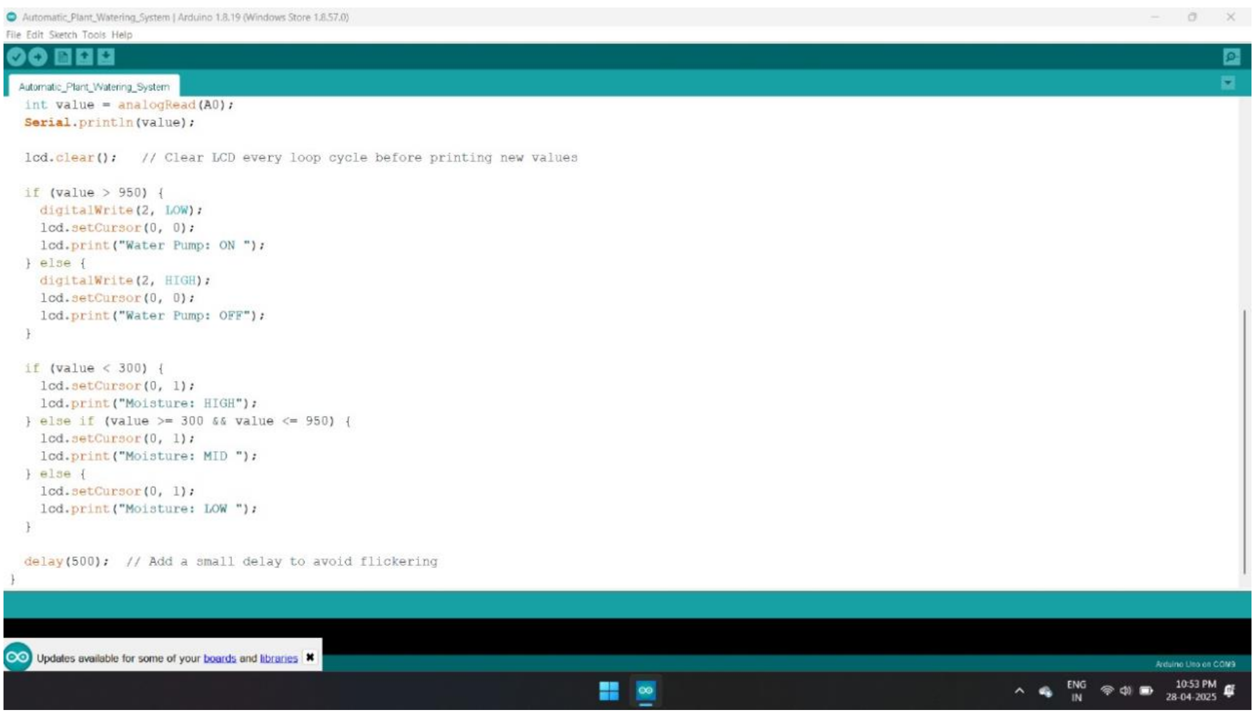
4. SOFTWARE DESCRIPTION

The code is for an automated system that uses a soil moisture sensor, a relay module, and a water pump to manage plant watering automatically. The soil moisture sensor continuously checks the soil's moisture level. When the sensor detects that the moisture is below a set threshold, the Arduino UNO processes the signal and activates the relay, which turns ON the water pump to irrigate the plant. While watering, the LCD screen (I2C) displays real-time messages like "Soil Dry - Watering..." and "Watering Completed" once the soil reaches the required moisture. If the soil is already moist, the pump remains OFF, and the LCD displays "Soil Moisture Normal".
This system ensures efficient plant care with minimal human intervention by using live soil feedback and smart control.

Source code in Arduino IDE software :



```
#include <LiquidCrystal_I2C.h>

LiquidCrystal_I2C lcd(0x3F, 16, 2);  // Use 0x3F or 0x27 based on your I2C address

void setup() {
  Serial.begin(9600);
  lcd.init();
  lcd.backlight();
  lcd.clear();
  pinMode(2, OUTPUT);
  digitalWrite(2, HIGH);

  lcd.setCursor(0, 0);
  lcd.print("IRRIGATION");
  lcd.setCursor(0, 1);
  lcd.print("SYSTEM IS ON ");
  delay(3000);
  lcd.clear();
}

void loop() {
  int value = analogRead(A0);
  Serial.println(value);

  lcd.clear();   // Clear LCD every loop cycle before printing new values

  if (value > 950) {
    digitalWrite(2, LOW);
```



```
  int value = analogRead(A0);
  Serial.println(value);

  lcd.clear();   // Clear LCD every loop cycle before printing new values

  if (value > 950) {
    digitalWrite(2, LOW);
    lcd.setCursor(0, 0);
    lcd.print("Water Pump: ON ");
  } else {
    digitalWrite(2, HIGH);
    lcd.setCursor(0, 0);
    lcd.print("Water Pump: OFF");
  }

  if (value < 300) {
    lcd.setCursor(0, 1);
    lcd.print("Moisture: HIGH");
  } else if (value >= 300 && value <= 950) {
    lcd.setCursor(0, 1);
    lcd.print("Moisture: MID ");
  } else {
    lcd.setCursor(0, 1);
    lcd.print("Moisture: LOW ");
  }

  delay(500);  // Add a small delay to avoid flickering
}
```

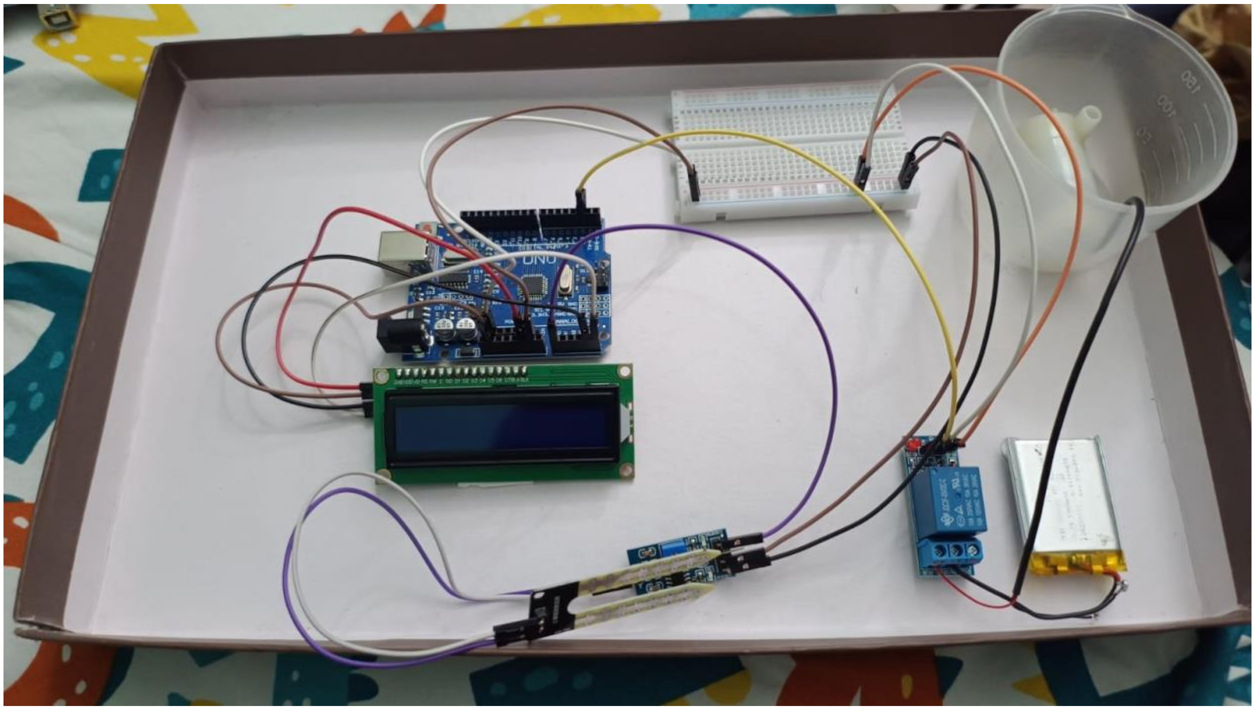
4.1 Source Code

```
#include <LiquidCrystal_I2C.h>
 LiquidCrystal_I2C lcd(0x3F, 16, 2);
 void setup() {
  Serial.begin(9600);
  lcd.init();
  lcd.backlight();
  lcd.clear();
  pinMode(2, OUTPUT);
  digitalWrite(2, HIGH);

  lcd.setCursor(0, 0);
  lcd.print("IRRIGATION");
  lcd.setCursor(0, 1);
  lcd.print("SYSTEM IS ON ");
  delay(3000);
  lcd.clear();
}

void loop() {
  int value = analogRead(A0);
  Serial.println(value);
   lcd.clear();
    if (value > 950) {
```

```
    digitalWrite(2, LOW);
    lcd.setCursor(0, 0);
    lcd.print("Water Pump: ON ");
  } else {
    digitalWrite(2, HIGH);
    lcd.setCursor(0, 0);
    lcd.print("Water Pump: OFF");
  }

  if (value < 300) {
    lcd.setCursor(0, 1);
    lcd.print("Moisture: HIGH");
  } else if (value >= 300 && value <= 950) {
    lcd.setCursor(0, 1);
    lcd.print("Moisture: MID ");
  } else {
    lcd.setCursor(0, 1);
    lcd.print("Moisture: LOW ");
  }
  delay(500);
}
```

5. RESULT AND DISCUSSION

The block diagram and circuit diagrams are shown once all the components are connected. All components such as the Arduino UNO, soil moisture sensor, relay, water pump, and LCD display are interconnected, forming the complete system setup. This setup helps in understanding the process flow in a simple and clear manner.



The Automated Plant Watering System operates based on real-time sensor inputs. Once the system is powered on after making the necessary connections and uploading the code to the Arduino, it uses the soil moisture sensor to constantly monitor the soil condition.

When the soil moisture sensor detects that the soil is dry (below the preset threshold), the system triggers the relay, which activates the water pump to irrigate the plant automatically. During the process, the LCD screen displays real-time updates like "Soil Dry - Watering..." or "Watering Completed" to keep the user informed.

Similar to the IR-based streetlight system which uses sensor inputs to control lighting, this plant watering system uses soil moisture readings to control the watering process. If the soil is already moist, the system keeps the pump OFF and displays "Soil Moisture Normal" on the LCD screen, ensuring water conservation and optimal plant care.

5.2 Result

The Automated Plant Watering System successfully operates as an efficient, user-friendly solution designed for smart and automatic plant care. By integrating a soil moisture sensor, relay module, water pump, and LCD display, the system accurately monitors the soil's moisture levels and responds accordingly.

When the soil moisture drops below a defined threshold, the system automatically activates the water pump to irrigate the plant. Once the desired moisture level is achieved, the pump turns off to prevent overwatering. Throughout the process, the LCD display provides clear instructions and real-time feedback, enhancing the overall user experience.

The project demonstrated the effectiveness of sensor-based automation in plant irrigation, ensuring optimal water usage and promoting healthy plant growth. It also highlighted how real-time monitoring and automatic control can reduce human intervention while maintaining efficiency. The system operates reliably with a simple and intuitive interface, making it a practical solution for home gardens, greenhouses, and agricultural applications.

6. CONCLUSION

The Automated Plant Watering System provides an efficient and smart solution for plant care, significantly reducing water wastage and manual effort. By utilizing sensor-based technology such as the soil moisture sensor and relay-controlled water pump, the system ensures that water is supplied only when necessary, promoting healthy plant growth and conserving valuable resources.

The automated operation of the system minimizes human intervention, reducing the chances of overwatering, underwatering, and associated errors. With the use of low-power components, the system operates continuously without significant power consumption, making it an eco-friendly and sustainable solution.

This project addresses the challenge of resource management by ensuring optimal water usage and limiting unnecessary energy use. While initial setup and hardware costs may be present, thoughtful design and high-quality components help reduce long-term maintenance needs, making the system a reliable and sustainable option for home gardens, greenhouses, and agricultural fields

7.REFERENCES

(I) https://www.electronicsforu.com/
(II) https://wokwi.com/
(III) https://www.tinkercad.com/